

**Victorian Certificate of Education**  
**2023**

**PHYSICS**

**Written examination**

**FORMULA SHEET**

**Instructions**

This formula sheet is provided for your reference.  
A question and answer book is provided with this formula sheet.

**Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.**

## Physics formulas

### Motion and related energy transformations

|                                                          |                                                                                                                |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| velocity; acceleration                                   | $v = \frac{\Delta s}{\Delta t}; \quad a = \frac{\Delta v}{\Delta t}$                                           |
| equations for constant acceleration                      | $v = u + at$ $s = ut + \frac{1}{2}at^2$ $s = vt - \frac{1}{2}at^2$ $v^2 = u^2 + 2as$ $s = \frac{1}{2}(v + u)t$ |
| Newton's second law                                      | $\Sigma F = ma$                                                                                                |
| circular motion                                          | $a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$                                                                     |
| Hooke's law                                              | $F = -k\Delta x$                                                                                               |
| elastic potential energy                                 | $\frac{1}{2}k(\Delta x)^2$                                                                                     |
| gravitational potential energy near the surface of Earth | $mg\Delta h$                                                                                                   |
| kinetic energy                                           | $\frac{1}{2}mv^2$                                                                                              |
| Newton's law of universal gravitation                    | $F = G \frac{m_1 m_2}{r^2}$                                                                                    |
| gravitational field                                      | $g = G \frac{M}{r^2}$                                                                                          |
| impulse                                                  | $F\Delta t$                                                                                                    |
| momentum                                                 | $mv$                                                                                                           |
| Lorentz factor                                           | $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$                                                                |
| time dilation                                            | $t = t_0 \gamma$                                                                                               |
| length contraction                                       | $L = \frac{L_0}{\gamma}$                                                                                       |
| rest energy                                              | $E_{\text{rest}} = mc^2$                                                                                       |
| relativistic total energy                                | $E_{\text{total}} = \gamma mc^2$                                                                               |
| relativistic kinetic energy                              | $E_k = (\gamma - 1)mc^2$                                                                                       |

## Fields and application of field concepts

|                                                        |                           |
|--------------------------------------------------------|---------------------------|
| electric field between charged plates                  | $E = \frac{V}{d}$         |
| energy transformations of charges in an electric field | $\frac{1}{2}mv^2 = qV$    |
| field of a point charge                                | $E = \frac{kq}{r^2}$      |
| force on an electric charge                            | $F = qE$                  |
| Coulomb's law                                          | $F = \frac{kq_1q_2}{r^2}$ |
| magnetic force on a moving charge                      | $F = qvB$                 |
| magnetic force on a current carrying conductor         | $F = nIlB$                |
| radius of a charged particle in a magnetic field       | $r = \frac{mv}{qB}$       |

## Generation and transmission of electricity

|                           |                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------|
| voltage; power            | $V = RI; \quad P = VI = I^2R$                                                                                 |
| resistors in series       | $R_T = R_1 + R_2$                                                                                             |
| resistors in parallel     | $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$                                                               |
| ideal transformer action  | $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$                                                         |
| AC voltage and current    | $V_{\text{RMS}} = \frac{1}{\sqrt{2}}V_{\text{peak}} \quad I_{\text{RMS}} = \frac{1}{\sqrt{2}}I_{\text{peak}}$ |
| electromagnetic induction | EMF: $\mathcal{E} = -N \frac{\Delta\Phi_B}{\Delta t}$ flux: $\Phi_B = B_{\perp}A$                             |
| transmission losses       | $V_{\text{drop}} = I_{\text{line}} R_{\text{line}} \quad P_{\text{loss}} = I_{\text{line}}^2 R_{\text{line}}$ |

## Wave concepts

|                                 |                                                         |
|---------------------------------|---------------------------------------------------------|
| wave equation                   | $v = f\lambda$                                          |
| constructive interference       | path difference $= n\lambda$                            |
| destructive interference        | path difference $= \left(n - \frac{1}{2}\right)\lambda$ |
| fringe spacing                  | $\Delta x = \frac{\lambda L}{d}$                        |
| Snell's law                     | $n_1 \sin\theta_1 = n_2 \sin\theta_2$                   |
| refractive index and wave speed | $n_1 v_1 = n_2 v_2$                                     |

**The nature of light and matter**

|                       |                         |
|-----------------------|-------------------------|
| photoelectric effect  | $E_{k\max} = hf - \phi$ |
| photon energy         | $E = hf$                |
| photon momentum       | $p = \frac{h}{\lambda}$ |
| de Broglie wavelength | $\lambda = \frac{h}{p}$ |

**Data**

|                                                |                                                                                |
|------------------------------------------------|--------------------------------------------------------------------------------|
| acceleration due to gravity at Earth's surface | $g = 9.8 \text{ m s}^{-2}$                                                     |
| mass of the electron                           | $m_e = 9.1 \times 10^{-31} \text{ kg}$                                         |
| magnitude of the charge of the electron        | $e = 1.6 \times 10^{-19} \text{ C}$                                            |
| Planck's constant                              | $h = 6.63 \times 10^{-34} \text{ J s}$ $h = 4.14 \times 10^{-15} \text{ eV s}$ |
| speed of light in a vacuum                     | $c = 3.0 \times 10^8 \text{ m s}^{-1}$                                         |
| universal gravitational constant               | $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$                       |
| mass of Earth                                  | $M_E = 5.98 \times 10^{24} \text{ kg}$                                         |
| radius of Earth                                | $R_E = 6.37 \times 10^6 \text{ m}$                                             |
| Coulomb constant                               | $k = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$                            |

**Prefixes/Units**

|                       |                      |                           |                               |
|-----------------------|----------------------|---------------------------|-------------------------------|
| p = pico = $10^{-12}$ | n = nano = $10^{-9}$ | $\mu$ = micro = $10^{-6}$ | m = milli = $10^{-3}$         |
| k = kilo = $10^3$     | M = mega = $10^6$    | G = giga = $10^9$         | t = tonne = $10^3 \text{ kg}$ |